The empirical recurrence for OEIS A063133, proved by transfer matrix

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8 September 2026

Abstract

OEIS A063133 carries an empirical recurrence of order 3 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to a bounded window of consecutive lines, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 8$ vertices and is C-finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The conjecture

OEIS A063133 is "Dimension of the space of weight 2n cusp forms for Gamma_0(65).". It has offset 1 and begins

$$5, 20, 32, 48, 60, 76, 88, 104, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> (the empirical recurrence quoted in the entry)

As of the "Last modified" line on the live entry (Aug 22 2019 12:18:12, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The condition the entry imposes is local: it constrains a bounded window of consecutive lines and nothing beyond. Take as vertices the admissible configurations of one such window and put an edge from u to v whenever v may follow u, which the condition decides by inspection of the two. An admissible array is then exactly a walk, so with M the adjacency matrix on $S = 8$ vertices and ι, τ the vectors recording which windows may begin and end an array,

$$a(n) \;=\; \iota^{\top} M^{n} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^3 - \sum_i c_i t^{3-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+3} - \sum_i c_i M^{j+3-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; a(n-1) + a(n-2) - a(n-3)$$

for every $n > 4$.

Proof. The vector $w = q(M)\tau$ was formed by 3 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 4 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 4$ the recurrence fails on the entry's own published terms, so no larger range holds. □

4 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 50 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A063133.

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[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.